LAB
Data Parallelism
CSCI 2911/2916

STEP 1: Pre-Activity Quiz to Assess Pre-Knowledge

Please go to the consent form and pre-lab quiz

Link provided

STEP TWO: Flag Maker Activity

- You’ll be pretending to be the computer drawing the flag of **Mauritius** (on the right).
- You’ll use paper and colored pencils but otherwise follow how a computer might draw it.
- You must fill in the squares.
- You must **color one square at a time.** Do not color in multiple squares at one time.

Img from: https://en.wikipedia.org/wiki/Mauritius

Flag Maker Activity

You must **color one square at a time.**

Do not color in multiple squares at one tir

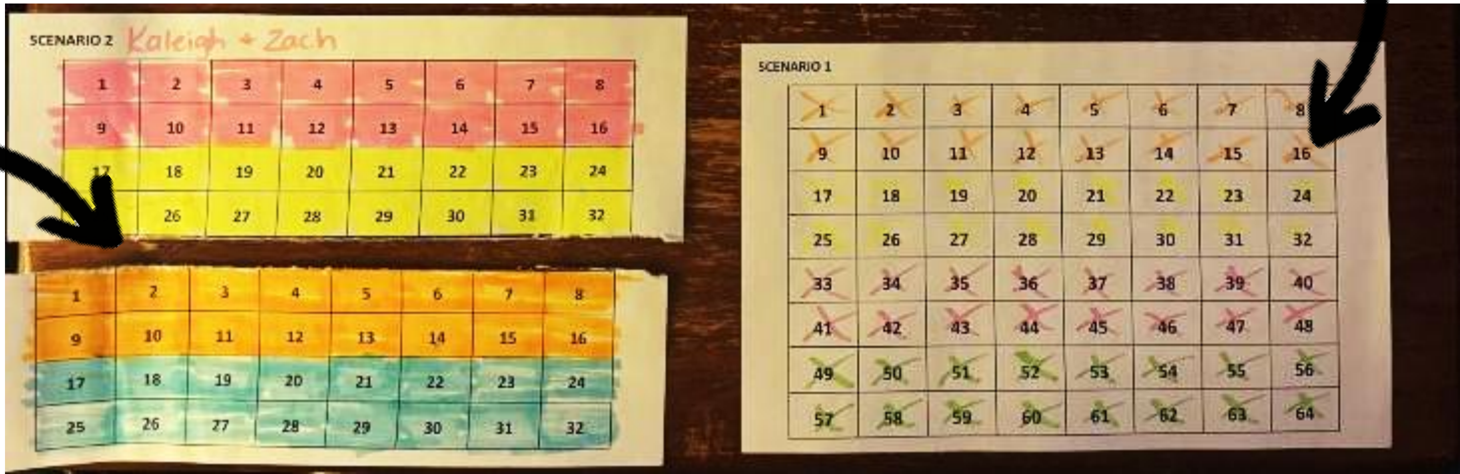
CORRECT!

WRONG!

Fill in the squares

WRONG!

Don't tear the paper!



Scenario 1: A Single Processor

Two people:

1. The Processor: colors first stripe (cell by cell), second stripe (cell by cell), third stripe (cell by cell), and then fourth stripe (cell by cell)
2. Timer: Times the Processor

P1	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32
	33	34	35	36	37	38	39	40
	41	42	43	44	45	46	47	48
	49	50	51	52	53	54	55	56
	57	58	59	60	61	62	63	64

Remember: squares must be mostly colored in & you can only color one square at a time!

Scenario 1: A Single Processor
QUESTIONS

1. Why does this process take a long time with only one “processor”?

2. How could this approach cause inefficiency in more complex tasks?

Scenario 2: Two Processors Collaborating

Three people:

1. Processor 1: Colors red and blue stripes (cell by cell)
2. Processor 2: Colors yellow and green stripes (cell by cell)
3. Timer: Times them; stop when **all** cells are colored.

P1	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32
P2	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
	17	18	19	20	21	22	23	24
	25	26	27	28	29	30	31	32

Remember: squares must be mostly colored in & you can only color one square at a time!

Scenario 2: Two Processors Collaborating
QUESTIONS

1. How does the addition of a second processor impact the overall completion time?

2. What potential problems could arise if both processors were given overlapping tasks such as coloring the same stripes?

Scenario 3: Four Processors Collaborating

Five people:

1. Processors 1-4: Each assigned to one color, which they color (cell by cell)
2. Timer: Times them; stop when **all** cells are colored.

P1	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P2	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P3	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16
P4	1	2	3	4	5	6	7	8
	9	10	11	12	13	14	15	16

Remember: squares must be mostly colored in & you can only color one square at a time!

Scenario 3: Four Processors Collaborating

QUESTIONS

1. How does using four processors further improve efficiency, and what challenges might this bring?

2. What would happen if one processor finished its task significantly earlier than the others?

Scenario 4: Four Processors Collaborating a Different Way (Column Assignments)

Five people:

1. Processors 1-4: Each assigned a strip of two columns (first takes left 2 columns, second takes next two columns, etc.) Processors colors these the correct color cell by cell.
2. Timer: Times them; stop when **all** cells are colored.

P1		P2		P3		P4	
1	9	1	9	1	9	1	9
2	10	2	10	2	10	2	10
3	11	3	11	3	11	3	11
4	12	4	12	4	12	4	12
5	13	5	13	5	13	5	13
6	14	6	14	6	14	6	14
7	15	7	15	7	15	7	15
8	16	8	16	8	16	8	16

Remember: squares must be mostly colored in & you can only color one square at a time!

Scenario 4: Four Processors Collaborating a Different Way (Column Assignments) QUESTIONS

1. How does assigning processors by columns rather than stripes impact performance and potential errors?

2. How does this scenario reflect real-world issues in distributed systems?

STEP 3: Post-Activity Quiz & Survey

Please take this post-lab quiz & Survey:

Post lab link provided

Survey link provided